

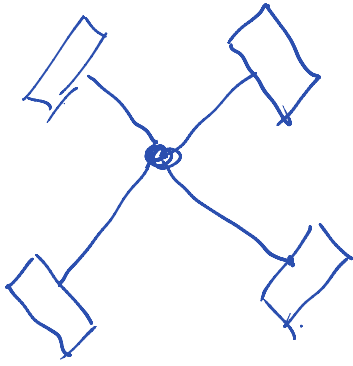
NOTEBOOK

麦克纳姆轮运动学解算



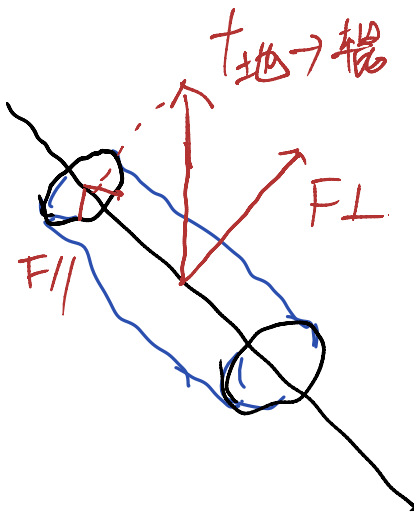
辋子与地面的接触

受力角度分析

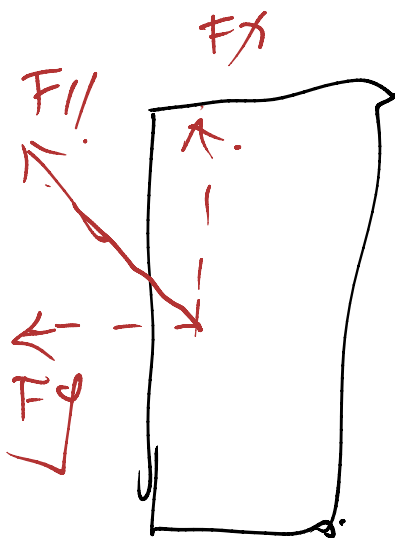


从力的角度进行分析

才能得到地面摩擦力对辋子的力



对单独的辋子受力分析



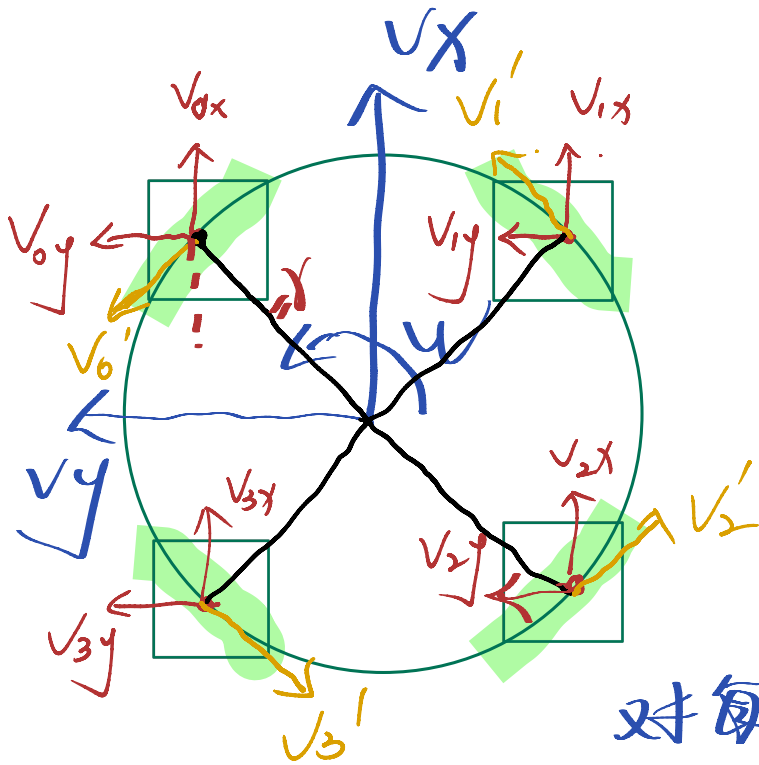
对整个辋子受力分析

由此可知每个辋子
由于地面摩擦可以
获得前向和侧向的力

从而很容易得到 整个小车速度的调整轮子
的速度方向

抵消 F_x 方向的力实现侧移

逆运动学分析



由于底盘是刚体
容易得到

$$V_x = V_{0x} = V_{1x} = V_{2x} = V_{3x}$$
$$V_y = V_{0y} = V_{1y} = V_{2y} = V_{3y}$$

设 V_n' 为对于车体 W
对每个轮子的速度影响

显然可得

$$W \cdot r = |V_0'| = |V_1'| = |V_2'| = |V_3'|$$

方向为 轮子中心与车体中心的连线
与圆的交点的切线方向

由此每个粒子的速度分量已经知道了

不妨进行向量合成

设每个粒子的速度为 V_n

则易得到

$$\begin{aligned}\vec{V}_0 &= \vec{V}_0x + \vec{V}_0y + \vec{V}_0' \\ &= \vec{V}_0x - V_0\cos 45^\circ + \vec{V}_0y + V_0'\cos 45^\circ\end{aligned}$$

为了方便起见设单位向量 $\begin{matrix} \uparrow i \\ j \end{matrix}$

$$\therefore \vec{V}_0 = (V_0x - V_0\cos 45^\circ)i +$$

$$(V_0y + V_0'\cos 45^\circ)j$$

$$\begin{aligned}\vec{V}_1 &= \vec{V}_1x + \vec{V}_1y + \vec{V}_1' \\ &= (V_1x + V_1'\cos 45^\circ)i + \\ &\quad + \\ &\quad (V_1y + V_1'\cos 45^\circ)j\end{aligned}$$

$$\begin{aligned}\vec{V}_2 &= \vec{V}_2 x + \vec{V}_2 y + \vec{V}_2' \\ &= (V_2 x + V_2' \cos 45^\circ) i \\ &\quad + (V_2 y - V_2' \cos 45^\circ) j\end{aligned}$$

$$\begin{aligned}\vec{V}_3 &= \vec{V}_3 x + \vec{V}_3 y + \vec{V}_3' \\ &= (V_3 x - V_3' \cos 45^\circ) i \\ &\quad + (V_3 y - V_3' \cos 45^\circ) j\end{aligned}$$

$$\therefore V_1 x = V_2 x = V_3 x = V_0 x = V_x$$

$$V_1 y = V_2 y = V_3 y = V_0 y = V_y$$

$$|V_0'| = |V_1'| = |V_2'| = |V_3'| = |\omega r|$$

轮子速度 V_n 与车体速度 V

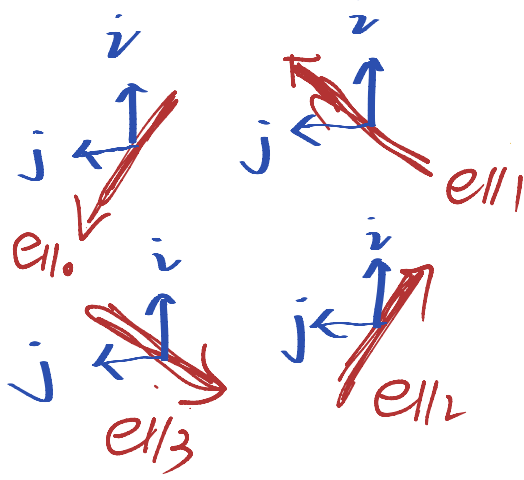
$$\therefore \vec{V}_0 = (V_x - \frac{r_2}{2} \omega r) i + (V_y + \frac{r_2}{2} \omega r) j$$

$$\vec{V}_1 = (V_x + \frac{r_2}{2} \omega r) i + (V_y + \frac{r_2}{2} \omega r) j$$

$$\vec{V}_2 = (V_x + \frac{r_2}{2} \omega r) i + (V_y - \frac{r_2}{2} \omega r) j$$

$$\vec{V}_3 = (V_x - \frac{r_2}{2} \omega r) i + (V_y - \frac{r_2}{2} \omega r) j$$

那轮子速度与电机转速关系呢?



$$\vec{V}_0 \rightarrow V_0 \omega$$

$$\vec{V}_1 \rightarrow \vec{V}_1 \omega$$

$$\vec{V}_2 \rightarrow \vec{V}_2 \omega$$

$$\vec{V}_3 \rightarrow \vec{V}_3 \omega$$

$$\vec{e}_{1/0} = -\frac{\pi_2}{2} \vec{i} + \frac{\pi_2}{2} \vec{j}$$

$$\vec{e}_{1/1} = \frac{\pi_2}{2} \vec{i} + \frac{\pi_2}{2} \vec{j}$$

$$\vec{e}_{1/2} = \frac{\pi_2}{2} \vec{i} - \frac{\pi_2}{2} \vec{j}$$

$$\vec{e}_{1/3} = -\frac{\pi_2}{2} \vec{i} - \frac{\pi_2}{2} \vec{j}$$

$$\vec{V}_0 = (V_x - \frac{\pi_2}{2} \omega r) \vec{i} + (V_y + \frac{\pi_2}{2} \omega r) \vec{j}$$

$$\vec{V}_1 = (V_x + \frac{\pi_2}{2} \omega r) \vec{i} + (V_y + \frac{\pi_2}{2} \omega r) \vec{j}$$

$$\vec{V}_2 = (V_x + \frac{\pi_2}{2} \omega r) \vec{i} + (V_y - \frac{\pi_2}{2} \omega r) \vec{j}$$

$$\vec{V}_3 = (V_x - \frac{\pi_2}{2} \omega r) \vec{i} + (V_y - \frac{\pi_2}{2} \omega r) \vec{j}$$

设每个轮子的速度 $\vec{V}_n$ 投影到轮子轴的速度为 $\vec{V}_{1/n}$

$$\begin{aligned} \text{则 } \vec{V}_{1/0} &= \vec{V}_0 \cdot \vec{e}_{1/0} = (V_x - \frac{\pi_2}{2} \omega r, V_y + \frac{\pi_2}{2} \omega r) \cdot (-\frac{\pi_2}{2}, \frac{\pi_2}{2}) \\ &= -\frac{\pi_2}{2} V_x + \frac{\pi_2}{2} V_y + \omega r \end{aligned}$$

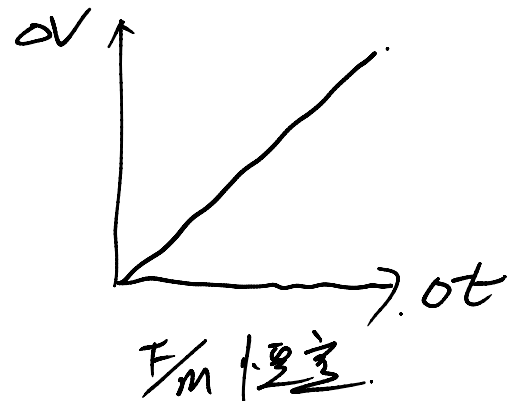
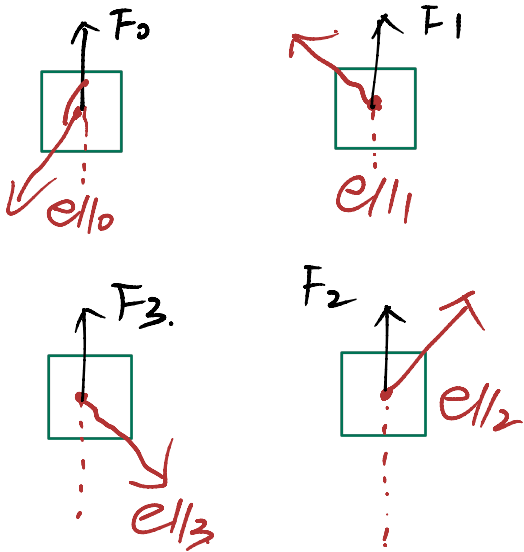
$$\begin{aligned} \vec{V}_{1/1} &= \vec{V}_1 \cdot \vec{e}_{1/1} = (V_x + \frac{\pi_2}{2} \omega r, V_y + \frac{\pi_2}{2} \omega r) \cdot (\frac{\pi_2}{2}, \frac{\pi_2}{2}) \\ &= \frac{\pi_2}{2} V_x + \frac{\pi_2}{2} V_y + \omega r \end{aligned}$$

→ → →

$$\begin{aligned} V_{1/2} &= V_2 \cdot e_{1/2} = (V_x + \frac{\sqrt{2}}{2} \omega r, V_y - \frac{\sqrt{2}}{2} \omega r) (\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}) \\ &= \frac{\sqrt{2}}{2} V_x - \frac{\sqrt{2}}{2} V_y + \omega r \end{aligned}$$

$$\begin{aligned} \vec{V}_{1/3} &= \vec{V}_3 - \vec{e}_{1/3} = (V_x - \frac{\sqrt{2}}{2} \omega r, V_y - \frac{\sqrt{2}}{2} \omega r) (-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}) \\ &= -\frac{\sqrt{2}}{2} V_x - \frac{\sqrt{2}}{2} V_y + \omega r \end{aligned}$$

那轮子轴向速度与轮子转速关系



↑表示轮子正转时
地面对轮子的摩擦力

$$\text{而 } F=ma \rightarrow F=m \frac{\Delta V}{\Delta t} \rightarrow F=\frac{m}{\Delta t} \Delta V$$

轮子的转速即电机的转速

与 F_n 成正比

轮 $\omega \uparrow$ $F_n \uparrow$

$$\text{设 } V_{\omega n} = \alpha F_n = \alpha \frac{m}{\Delta t} \Delta V_n$$

$$\text{设 } k = \alpha \frac{m}{\Delta t} \quad \text{则 } V_{\omega n} = k \Delta V_n$$

不过这里的 ΔV 的方向与 F_n 的方向是一致的

但是我们要得到是 V_w 与 $V_{//}$ 之间的关系.

向量分解易得.

$$\Delta V_0 \cos 135^\circ = \Delta V_{//0}$$

$$\Delta V_3 \cos 135^\circ = \Delta V_{//3}$$

$$\text{for } V_{wn} = k \Delta V_n$$

$$\frac{1}{k} V_{w0} \cos 135^\circ = V_{//0}$$

$$\frac{1}{k} V_{w3} \cos 135^\circ = V_{//3}$$

$$\Delta V_1 \cos 45^\circ = \Delta V_{//1}$$

$$\Delta V_2 \cos 45^\circ = \Delta V_{//2}$$

$$\frac{1}{k} V_{w1} \cos 45^\circ = V_{//1}$$

$$\frac{1}{k} V_{w2} \cos 45^\circ = V_{//2}$$

而我们上一步已经得到 $\theta_{//n}$ 方向的目标速度 $V_{//n}$

此外 $\frac{1}{k}$ 暂且取1 此外全向轮处原因一致.

$$\vec{V}_{//0} = \left(-\frac{\sqrt{2}}{2} V_x + \frac{\sqrt{2}}{2} V_y + W_r \right) = V_{w0} \cos 135^\circ = V_{w0} \left(-\frac{\sqrt{2}}{2} \right)$$

$$\vec{V}_{//1} = \left(\frac{\sqrt{2}}{2} V_x + \frac{\sqrt{2}}{2} V_y + W_r \right) = V_{w1} \cos 45^\circ = V_{w1} \frac{\sqrt{2}}{2}$$

$$\vec{V}_{//2} = \left(\frac{\sqrt{2}}{2} V_x - \frac{\sqrt{2}}{2} V_y + W_r \right) = V_{w2} \cos 45^\circ = V_{w2} \frac{\sqrt{2}}{2}$$

$$\vec{V}_{//3} = \left(-\frac{\sqrt{2}}{2} V_x - \frac{\sqrt{2}}{2} V_y + W_r \right) = V_{w3} \cos 135^\circ = V_{w3} \left(-\frac{\sqrt{2}}{2} \right)$$

$$\therefore V_{w0} = V_x - V_y - \sqrt{2} W_r$$

$$V_{w1} = V_x + V_y + \sqrt{2} W_r$$

$$V_{w2} = V_x - V_y + \sqrt{2} W_r$$

$$V_{w3} = V_x + V_y - \sqrt{2} W_r$$

考虑到电机安装. 另一侧电机则须方向相反.

V_{w0} V_{w3} 即 0 3 电机位于左侧

V_{w1} V_{w2} 即 1 2 电机位于右侧

以 12 为正则 03 须反

V_x V_y w

$$\begin{aligned} \therefore V_{w0} &= -V_x + V_y + \sqrt{2}w \\ V_{w1} &= V_x + V_y + \sqrt{2}w \\ V_{w2} &= V_x - V_y + \sqrt{2}w \\ V_{w3} &= -V_x - V_y + \sqrt{2}w \end{aligned} \quad \begin{bmatrix} V_{w0} \\ V_{w1} \\ V_{w2} \\ V_{w3} \end{bmatrix} = \begin{bmatrix} -1 & 1 & \sqrt{2} \\ 1 & 1 & \sqrt{2} \\ 1 & -1 & \sqrt{2} \\ -1 & -1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ w \end{bmatrix}$$

又考虑到电机正电流旋转方向

取负.

V_x V_y w

$$\begin{bmatrix} V_{w0} \\ V_{w1} \\ V_{w2} \\ V_{w3} \end{bmatrix} = \begin{bmatrix} 1 & -1 & -\sqrt{2} \\ -1 & -1 & -\sqrt{2} \\ -1 & +1 & -\sqrt{2} \\ 1 & 1 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ w \end{bmatrix}$$

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wheel_speed[0] = vx_set - vy_set - MOTOR_DISTANCE_TO_CENTER * wz_set;
wheel_speed[1] = -vx_set - vy_set - MOTOR_DISTANCE_TO_CENTER * wz_set;
wheel_speed[2] = -vx_set + vy_set - MOTOR_DISTANCE_TO_CENTER * wz_set;
wheel_speed[3] = vx_set + vy_set - MOTOR_DISTANCE_TO_CENTER * wz_set;

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